Avijit Podder, PhD

Computational Data Scientist | Multi-Omics & Precision Medicine

University of Arizona College of Medicine - Tucson, Arizona

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RESEARCH PROFILE

Computational biologist and data scientist with more than 15 years of experience in multi-omics integration, network biology, genomics, and computational therapeutics. Research focuses on individualized molecular profiling and precision medicine, with applications in Alzheimer's disease, Parkinson's disease, aging, cross-species biology, and other complex disorders. Experienced in leading interdisciplinary analyses that integrate proteomic, transcriptomic, genomic, pathway, network, and clinical data to identify disease mechanisms, molecular subgroups, biomarkers, and therapeutic hypotheses.

RESEARCH INTERESTS

- Individualized and single-subject multi-omics
- Neurodegenerative disease (Alzheimer's disease and Parkinson's disease)
- Cross-species genomics and translational biology
- Statistical modeling, machine learning, and reproducible biomedical data science
- Proteomics and transcriptomics
- Network biology, pathway modeling, and patient similarity
- Precision medicine, biomarker discovery, and computational therapeutics

PROFESSIONAL EXPERIENCE

2026-Present — **Computational Data Scientist** | **University of Arizona College of Medicine - Tucson**

Computational multi-omics, precision medicine, and translational neurodegeneration research.

2023-2026 — **Associate Computational Scientist** | **The Jackson Laboratory for Genomic Medicine, Farmington, CT**

Designed personalized therapeutic strategies using individual transcriptomic and proteomic signatures in Alzheimer's disease; performed cross-species multi-omics and network-based functional analyses to identify disease-relevant pathways and targets.

2019-2023 — **Postdoctoral Fellow** | **Translational Genomics Research Institute (TGen), Phoenix, AZ**

Conducted cross-species network and pathway analyses in aging and longevity; developed personalized pathway workflows for cancer; designed enrichment and multi-omics workflows for organoid, iPSC, scRNA-seq, and genomic datasets.

2012-2019 — **Senior Research Fellow** | **Sri Venkateswara College, University of Delhi, India**

Applied systems biology, protein structure modeling, molecular dynamics, docking, and computer-aided drug design to neurological disorders including schizophrenia and Alzheimer's disease.

2011-2012 — **Senior Research Fellow** | **ICAR-National Bureau of Animal Genetic Resources, Karnal, India**

Performed population genomics, SNP analysis, haplotype analysis, and comparative pathway/network analyses in livestock genomics.

2009-2011 — **Project Assistant** | **CSIR-Institute of Genomics and Integrative Biology, Delhi, India**

Analyzed genome-wide genotyping, imputation, sequencing, and functional annotation data in type 2 diabetes and metabolic syndrome research.

EDUCATION

PhD, Computational Biology | 2014-2019

University of Delhi South Campus, Delhi, India

Dissertation: Computational Studies of Drug Targets in Dopaminergic System Implicated in Neurological Disorders

Master of Science & Technology, Bioinformatics | 2006-2009

West Bengal University of Technology, Kolkata, India

BSc, Zoology | 2002-2005

SELECTED PEER-REVIEWED PUBLICATIONS

2026. Podder A, Liew YJ, Cary GA, Carter GW, Uyar A. **Single-subject Proteomic Signatures in Alzheimer's Disease Reflect Clinical Phenotypes and Distinguish Asymptomatic from Symptomatic Cases.** Alzheimer's & Dementia: Translational Research & Clinical Interventions. DOI

2023. Frankel E, Podder A, Sharifi M, Pillai R, Belnap N, et al. **Genetic and Protein Network Underlying the Convergence of Rett-Syndrome-like (RTT-L) Phenotype in Neurodevelopmental Disorders.** Cells 12(10):1437. DOI

2021. Podder A, Raju A, Schork NJ. **Cross-Species and Human Inter-Tissue Network Analysis of Genes Implicated in Longevity and Aging Reveal Strong Support for Nutrient Sensing.** Frontiers in Genetics 12:719713. DOI

2018. Podder A, Pandit M, Latha N. **Drug Target Prioritization for Alzheimer's Disease Using Protein Interaction Network Analysis.** OMICS: A Journal of Integrative Biology 22(10):665-677. DOI

2017. Podder A, Latha N. **Data on Overlapping Brain Disorders and Emerging Drug Targets in Human Dopamine Receptors Interactions Network.** Data in Brief 12:277-286.

2016. Pandey D, Podder A, Pandit M, Latha N. **CD4-gp120 Interaction Interface - A Gateway for HIV-1 Infection in Human: Molecular Network, Modeling and Docking Studies.** Journal of Biomolecular Structure & Dynamics 35(12):2631-2644.

2016. Podder A, Pandey D, Latha N. **Investigating the structural impact of S311C mutation in DRD2 receptor by molecular dynamics & docking studies.** Biochimie 123:52-64. DOI

2014. Podder A, Latha N. **New Insights into Schizophrenia Disease Genes Interactome in the Human Brain: Emerging Targets and Therapeutic Implications in the Postgenomics Era.** OMICS: A Journal of Integrative Biology 18(12):754-766.

2014. Podder A, Jatana N, Latha N. **Human Dopamine Receptors Interaction Network (DRIN): A systems biology perspective on topology, stability and functionality of the network.** Journal of Theoretical Biology 357:169-183.

2012. Jaiswal A, Tabassum R, Podder A, Ghosh S, Tandon N, Bharadwaj D. **Elevated level of C-reactive protein is associated with risk of Prediabetes in Indians.** Atherosclerosis 222(2):495-501.

2012. Mahajan A, Jaiswal A, Tabassum R, Podder A, Ghosh S, et al. **Elevated levels of C-reactive protein as a risk factor for Metabolic Syndrome in Indians.** Atherosclerosis 220(1):275-281.

SELECTED PRESENTATIONS

2025 - Poster - Alzheimer's Association International Conference (AAIC), Toronto, Canada: Harnessing Individual Omics Signatures for Precision Medicine in Alzheimer's Disease

2024 - Poster - American Society of Human Genetics (ASHG), Denver, CO: Single-subject omics approaches unravel pathway-specific heterogeneity in Alzheimer's Disease

2021 - Poster - American Society of Human Genetics (ASHG), Virtual: Implications of biological networks: Search for the basal gene network using plasma proteomic signatures of COVID-19 patients

2020 - Poster - American Society of Human Genetics (ASHG), Virtual: Cross-species network analysis of genes implicated in human aging reveal strong differences in nutrient sensing

2018 - Oral presentation - 19th International Conference on Systems Biology, Lyon, France: Computational Studies of Drug Targets for Human Dopaminergic System Implicated in Neurological Disorders

INTELLECTUAL PROPERTY

Patent pending: Nicholas J. Schork and Avijit Podder. Genomic Prediction of Discordant Gene-Phenotype Relationships Between Species. U.S. Patent Application No. 63/592,554, filed October 23, 2023.

EDITORIAL & PEER-REVIEW SERVICE

2025-Present | Academic Reviewer - Neurochemical Research, Springer Nature

2025-Present | Academic Reviewer - Discover Applied Sciences, Springer Nature

2023-Present | Review Editor - Frontiers in Systems Biology

2023-Present | Review Editor - Frontiers in Bioinformatics - Network Bioinformatics

2019-Present | Academic Reviewer - Nature Scientific Reports

2018-Present | Editorial Board Member - Journal of Biomolecular Research & Therapeutics

TEACHING & MENTORSHIP

2020-2022 - Guided undergraduate summer trainees through the Helios Scholars program at TGen; supported development of an R Shiny application and research outputs.

2016-2018 - Guided undergraduate trainees in network analysis and systems biology of neurological disorders through the DU Star Innovative Project.

2014-2017 - Organized workshops for college teachers and postgraduate students on molecular modeling and drug design.

TECHNICAL EXPERTISE

Domain	Expertise
Omics & Bioinformatics	Multi-omics integration; proteomics; transcriptomics/RNA-seq; genomics; scRNA-seq; functional enrichment; comparative genomics
Systems & Network Biology	Protein-protein interaction networks; graph theory; pathway analysis; Cytoscape; patient-similarity approaches
Programming & Statistics	R/Bioconductor; Python; Bash/Linux; statistical modeling; reproducible workflows; scientific visualization
Computational Therapeutics	Protein structure modeling; molecular dynamics; docking; mutation effect analysis; computer-aided drug design